

Lab 05 - ME315

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```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
```

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-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
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```
library(lubridate)
```

```
library(leaflet)
```

```
analisa_aeronave <- function(
  tail_number,
  arquivo,
  arquivo_airports = "airports.csv") {
```

```
  dados_aeroportos <- read_csv(arquivo_airports, col_types = cols(
    IATA_CODE = col_character(),
    AIRPORT = col_character(),
    LATITUDE = col_double(),
    LONGITUDE = col_double()), show_col_types = FALSE) %>%
  select(IATA_CODE, AIRPORT, LATITUDE, LONGITUDE)
```

```
voos_filtrados <- read_csv_chunked(
  file = arquivo,
  callback = DataFrameCallback$new(
```

```

function(chunk, index) {
  chunk %>%
    filter(TAIL_NUMBER == !!tail_number)}),
chunk_size = 100000,
col_types = cols(
  YEAR = col_double(),
  MONTH = col_double(),
  DAY = col_double(),
  SCHEDULED_DEPARTURE = col_double(),
  TAIL_NUMBER = col_character(),
  ORIGIN_AIRPORT = col_character(),
  DESTINATION_AIRPORT = col_character(),
  DISTANCE = col_double(),
  AIR_TIME = col_double()), show_col_types = FALSE)

if (nrow(voos_filtrados) == 0) {
  stop(paste0("Nenhum voo encontrado para o TAIL_NUMBER '",
    tail_number, "'."))}

tabela_trajetos <- voos_filtrados %>%
  mutate(
    horario_num = as.integer(SCHEDULED_DEPARTURE),
    data_hora_decolagem = make_datetime(
      year = as.integer(YEAR),
      month = as.integer(MONTH),
      day = as.integer(DAY),
      hour = horario_num %/% 100,
      min = horario_num %% 100)) %>%
  arrange(data_hora_decolagem) %>%
  left_join(dados_aeroportos, by = c("ORIGIN_AIRPORT" = "IATA_CODE")) %>%
  rename(nome_origem = AIRPORT, lat_origem = LATITUDE, lon_origem = LONGITUDE) %>%
  left_join(dados_aeroportos, by = c("DESTINATION_AIRPORT" = "IATA_CODE")) %>%
  rename(nome_destino = AIRPORT, lat_destino = LATITUDE, lon_destino = LONGITUDE) %>%
  mutate(
    velocidade_media = if_else(
      !is.na(AIR_TIME) & AIR_TIME > 0,
      (DISTANCE / AIR_TIME) * 60,
      NA_real_))

sequencia_trajeto <- c(
  tabela_trajetos$ORIGIN_AIRPORT[1],
  tabela_trajetos$DESTINATION_AIRPORT)

```

```

sequencia_trajeto <- paste(
  sequencia_trajeto,
  collapse = " → ")

velocidades_validas <- tabela_trajetos$velocidade_media[
  !is.na(tabela_trajetos$velocidade_media)]

if (length(velocidades_validas) == 0) {
  tabela_trajetos <- tabela_trajetos %>%
    mutate(espessura_linha = 2)

} else if (length(unique(velocidades_validas)) == 1) {

  tabela_trajetos <- tabela_trajetos %>%
    mutate(espessura_linha = if_else(is.na(velocidade_media), 2, 3))

} else {

  tabela_trajetos <- tabela_trajetos %>%
    mutate(
      espessura_linha = if_else(
        is.na(velocidade_media),
        2,
        scales::rescale(
          velocidade_media,
          to = c(1, 6),
          from = range(
            velocidade_media,
            na.rm = TRUE))))))}

dados_mapa <- tabela_trajetos %>%
  filter(!is.na(lat_origem),
         !is.na(lon_origem),
         !is.na(lat_destino),
         !is.na(lon_destino))

aeroportos_visitados <- bind_rows(
  tabela_trajetos %>%
    transmute(
      IATA = ORIGIN_AIRPORT,
      nome = nome_origem,
      lat = lat_origem,

```

```

lon = lon_origem), tabela_trajetos %>%
transmute(
  IATA = DESTINATION_AIRPORT,
  nome = nome_destino,
  lat = lat_destino,
  lon = lon_destino)) %>%
filter(!is.na(lat), !is.na(lon)) %>%
distinct(IATA, .keep_all = TRUE)

numero_voos <- nrow(tabela_trajetos)

numero_aeroportos <- nrow(aeroportos_visitados)

velocidade_media_geral <- mean(
  tabela_trajetos$velocidade_media,
  na.rm = TRUE)

if (is.nan(velocidade_media_geral)) {
  velocidade_media_geral <- NA_real_}

mapa_trajetos <- leaflet() %>%
  addTiles()

if (nrow(dados_mapa) > 0) {

  for (i in seq_len(nrow(dados_mapa))) {

    voo <- dados_mapa[i, ]

    pop_info <- paste0(
      "<b>Aeronave:</b> ", tail_number,
      "<br>",
      "<b>Data:</b> ",
      format(
        voo$data_hora_decolagem,
        "%d/%m/%Y %H:%M"
      ),
      "<br>",
      "<b>Origem:</b> ",
      voo$ORIGIN_AIRPORT,
      " - ",
      voo$nome_origem,

```

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"<br>",
"<b>Destino:</b> ",
voo$DESTINATION_AIRPORT,
" - ",
voo$nome_destino,
"<br>",
"<b>Distância:</b> ",
round(voo$DISTANCE, 1),
" milhas",
"<br>",
"<b>Tempo de voo:</b> ",
ifelse(
  is.na(voo$AIR_TIME),
  "N/A",
  paste0(
    round(voo$AIR_TIME, 0),
    " minutos")),
"<br>",
"<b>Velocidade média:</b> ",
ifelse(
  is.na(voo$velocidade_media),
  "N/A",
  paste0(
    round(voo$velocidade_media, 1),
    " mph")))

mapa_trajetos <- mapa_trajetos %>%
  addPolylines(
    lng = c(
      voo$lon_origem,
      voo$lon_destino),
    lat = c(
      voo$lat_origem,
      voo$lat_destino),
    weight = voo$espessura_linha,
    opacity = 0.7,
    color = "blue",
    popup = pop_info)}}

if (nrow(aeroportos_visitados) > 0) {

  mapa_trajetos <- mapa_trajetos %>%

```

```

addCircleMarkers(
  data = aeroportos_visitados,
  lng = ~lon,
  lat = ~lat,
  radius = 5,
  color = "red",
  fillOpacity = 0.8,
  popup = ~paste0(
    "<b>",
    IATA,
    "</b>",
    "<br>",
    nome), label = ~IATA)}

velocidade_texto <- ifelse(
  is.na(velocidade_media_geral),
  "N/A",
  paste0(
    round(velocidade_media_geral, 1),
    " mph"))

estatisticas <- paste0(
  "<div style='
    background-color: white;
    padding: 10px;
    border-radius: 5px;
    box-shadow: 0 1px 5px rgba(0,0,0,0.4);
    font-size: 13px;
  '>",
  "<b>Aeronave:</b> ",
  tail_number,
  "<br>",
  "<b>Número de voos:</b> ",
  numero_voos,
  "<br>",
  "<b>Aeroportos visitados:</b> ",
  numero_aeroportos,
  "<br>",
  "<b>Velocidade média:</b> ",
  velocidade_texto,
  "<br>",
  "<b>Trajeto:</b> ",

```

```

sequencia_trajeto,
"</div>")

mapa_trajetos <- mapa_trajetos %>%
  addControl(
    html = estatisticas,
    position = "topright")

return(
  list(
    tabela = tabela_trajetos,
    mapa = mapa_trajetos)))

resultado <- analisa_aeronave(
  tail_number = "N431WN",
  arquivo = "flights.csv.zip",
  arquivo_airports = "airports.csv"
)

resultado$tabela

```

```

# A tibble: 1,938 x 41
  YEAR MONTH DAY DAY_OF_WEEK AIRLINE FLIGHT_NUMBER TAIL_NUMBER
  <dbl> <dbl> <dbl>         <dbl> <chr>          <dbl> <chr>
1  2015     1     1             4 WN             1762 N431WN
2  2015     1     1             4 WN             3564 N431WN
3  2015     1     1             4 WN             3564 N431WN
4  2015     1     1             4 WN             3564 N431WN
5  2015     1     2             5 WN             1793 N431WN
6  2015     1     2             5 WN             1793 N431WN
7  2015     1     2             5 WN             1793 N431WN
8  2015     1     2             5 WN             1793 N431WN
9  2015     1     2             5 WN             1793 N431WN
10 2015     1     2             5 WN             3144 N431WN
# i 1,928 more rows
# i 34 more variables: ORIGIN_AIRPORT <chr>, DESTINATION_AIRPORT <chr>,
# SCHEDULED_DEPARTURE <dbl>, DEPARTURE_TIME <chr>, DEPARTURE_DELAY <dbl>,
# TAXI_OUT <dbl>, WHEELS_OFF <chr>, SCHEDULED_TIME <dbl>, ELAPSED_TIME <dbl>,
# AIR_TIME <dbl>, DISTANCE <dbl>, WHEELS_ON <chr>, TAXI_IN <dbl>,
# SCHEDULED_ARRIVAL <chr>, ARRIVAL_TIME <chr>, ARRIVAL_DELAY <dbl>,
# DIVERTED <dbl>, CANCELLED <dbl>, CANCELLATION_REASON <chr>, ...

```

```
resultado$mapa
```

Aeronave: N431WN

Número de voos: 1938

Aeroportos visitados: 81

Velocidade média: 418.2 mph

Trajetos: DCA → HOU → BOS → BWI → MCI → LAX → LAS → ICT → DAL → MDW → ATL → MCO → BWI → MDW → AUS → MDW → BWI → BNA → MDW → GSP → MDW → DCA → ATL → RSW → MKE → BWI → MCO → CAK → BOS → MDW → PHX → SFO → ATL → LGA → CAK → TPA → RDU → ATL → LAS → LAX → PHX → SNA → SJC → LAX → DEN → MSP → MDW → CMH → BWI → DEN → STL → DCA → STL → MCO → PIT → PHX → SFO → SNA → SJC → DEN → MDW → ROC → BWI → BHM → DAL → SFO → MKE → DCA → STL → MKE → LAX → OAK → DEN → MSY → DEN → SAN → MDW → CMH → STL → DAL → SAN → DEN → EWR → MDW → HOU → AUS → DCA → MDW → CLT → MDW → SAN → OAK → PDX → PHX → BNA → MCO → DAY → BWI → BDL → BWI → RSW → STL → LAX → OAK → LAX → DEN → IND → FLL → ISP → MCO → BNA → DEN → SEA → DEN → IAD → SAN → SJC → SNA → SJC → SNA → SMF → LAS → AUS → BWI → LAS → OAK → BUR → LAS → MDW → PDX → SJC → MDW → PIT → BWI → JAX → MDW → BNA → MDW → ATL → HOU → SNA → EWR → LAS → PHX → STL → BUF → MDW → BNA → MDW → DCA → MDW → MCO → PVD → TPA → BNA → LGA → HOU → ATL → DEN → PDX → OAK → SAN → PHX → ELP → LAX → TUS → SAN → OAK → DAL → DCA → BNA → DEN → AUS → DEN → BNA → MSY → LAX → LAS → SAN → SJC → SAN → PDX → MCI → DCA → BNA → HOU → OAK → ONT → SMF → BUR → LAS → DSM → MDW → STL → BWI → SDF → LAS → OAK → SNA → SJC → BUR → LAS → MCI → BWI → BDL → RSW → PIT → RSW → ATL → BDL → ATL → MDW → SLC → OAK → BUR → OAK → SAN → LAS → DAL → DEN → LAS → BUR → OAK → LAS → ABQ → DAL → AUS → LAS → SJC → LAS → MKE → BWI → DEN → SNA → SJC → LAX → HOU → EWR → DEN → TPA → HOU → MDW → MCO → BDL → BWI → BNA → MCO → LAS → MCO → AUS → MCO → ALB → BWI → FLL → DEN → OAK → MDW → DSM → MDW → CMH → TPA → FLL → BUF → MCO → HOU → CHS → MDW → DTW → PHX → SLC → LAS → BHM → DAL → SAN → SMF → SAN → PHX → IND → HOU → LAX → LAS → OAK → GEG → LAS → ONT → SJC → LAS → LAX → BNA → DAL → DEN → ONT → PHX → SAN → SJC → SAN → AUS → HOU → BOS → CAK → DEN → LAS → SLC → LAS → BHM → DAL → SAN → SMF → MDW → BNA → PHX → OAK → LAS → OAK → SNA → DAL → STL → MDW → DCA → MKE → RSW → BDL → MDW → HOU → DAL → MCI → MDW → MSP → MCI → BWI → ELP → DAY → BWI → TPA → HOU → DAL → HOU → PHX → LAX → OAK → GEG → LAS → ATL → DCA → STL → MDW → RSW → GRR → DEN → IND → DEN → OAK → PHX → BOI → LAS → DEN → PHX → CMH → LAS → DEN → LAS → BUR → LAS → DEN → MCO → SJU → FLL → BUF → BWI → MSY → FLL → BWI → PVD → MCO → MKE → DCA → IND → DEN → SFO → SNA → OAK → SNA → SJC → LAS → BUR → OAK → LAX → SLC → LAX → PHX → MCO → MDW → RSW → MDW → HOU → DEN → LAS → SAT → BNA → DEN → SJC → SNA → OAK → LAX → MCI → PHX → SMF → PDX → MDW → MCI → ATL → CMH → MCO → AUS → DEN → MDW → MSP → MDW → MCO → ATL → RDU → ATL → LAX → SMF → SAN → MCO → MKE → DEN → OAK → BUR → SMF → ONT → PHX → MCI → PHX → SMF → PDX → MDW → MCI → STL → DTW → MDW → AUS → ATL → JAX → DEN → OAK → SAN → SMF → BUR → SJC → SNA → DAL → DEN → SEA → LAS → SNA → SJC → SNA → PHX → RDU → MCO → PHL → ATL → FLL → IND → HOU → LAX → ATL → TPA → BHM → LAS → STL → BWI → PVD → MCO → ISP → BWI → JAX → DEN → OAK → SAN → SMF → BUR → SJC → SNA → OAK → SEA → LAS → ATL → CAK → LGA → MKE → LAX → SJC → BUR → OAK → LAX → SJC → LAX → PHX → HOU → DCA → ATL → DEN → PDX → MDW → MCO → MEM → HOU → OAK → LAS → PHX → ABQ → MDW → FLL → MSY → TPA → PVD → MCO → ALB → MDW → SLC → LAX → OAK → PHX → OAK → BUR → LAS → MDW → RSW → MKE → PHX → PDX → SAN → LAS → JAX → HOU → SAT → LAS → LAX → PHX → TPA → MCI → LAS → SMF → PHX → LAS → ATL → MDW → LAS → PHX → MDW → PHX → BUR → SJC → BUR → SJC → SAN → BNA → AUS → DEN → PDX → LAS → BOI → LAS → SJC → LAX → SJC → SAN → OAK → MDW → LAS → BNA → MDW → TUL → MDW → MSP → STL → HOU → DAL → STL → DAL → MDW → LGA → HOU → BOS → HOU → BNA → DCA → CMH → ATL → FLL → PHL → MDW → DTW → MDW → DAL → DEN → SNA → SMF → PDX → MDW → CLT → MDW → SAT → SJC → ONT → SMF → LAS → SAN → DEN → IND → BWI → SDF → MDW → MCO → CAK → LGA → ATL → PIT → PHX → ABQ → DAL → STL → EWR → MSY → BWI → FLL → MCI → DEN → PDX → OAK → ABQ → OAK → BUR → SMF → LAX → STL → DAL → FLL → ISP → PBI → ISP → MCO → BHM → MDW → STL → ECP → DAL → LAS → SFO → SNA → LAS → RNO → LAS → SAN → PHX → MDW → HOU → EWR → MSY → BWI → FLL → MCI → DEN → PDX → OAK → ABQ → OAK → BUR → SMF → LAX → STL → DAL → FLL → DEN → SAT → MS

Leaflet | © OpenStreetMap, ODbL

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-15 21:45:49 -03"
```